

StratRoom Intelligence Hub

Retrieval-Augmented Generation (RAG) Architecture & System Specification

Scope: Enterprise Document RAG

Engine: Dual-Mode Hybrid Search

Security: Multi-Tenant & RBAC

Status: Production Spec

1. Executive Summary & Design Architecture

StratRoom's Intelligence Hub introduces a **Retrieval-Augmented Generation (RAG) Architecture** designed specifically for enterprise governance, risk, compliance, and strategic decision documentation. While StratRoom's core operational system utilizes structured tool-calling for relational MySQL tables, the Intelligence Hub handles **unstructured knowledge sources**—including policy frameworks, board meeting decks, regulatory filings, vendor contracts, and audit reports.

Core Architectural Principle: Hybrid Intelligence

RAG operates in tandem with StratRoom's SQL Agent Runner. Structured queries (KPI values, task counts, budget lines) execute direct MySQL queries, while qualitative and semantic inquiries ("What are our SLA default penalties in Q3 contracts?") route through the Vector RAG Engine.

Primary RAG Objectives

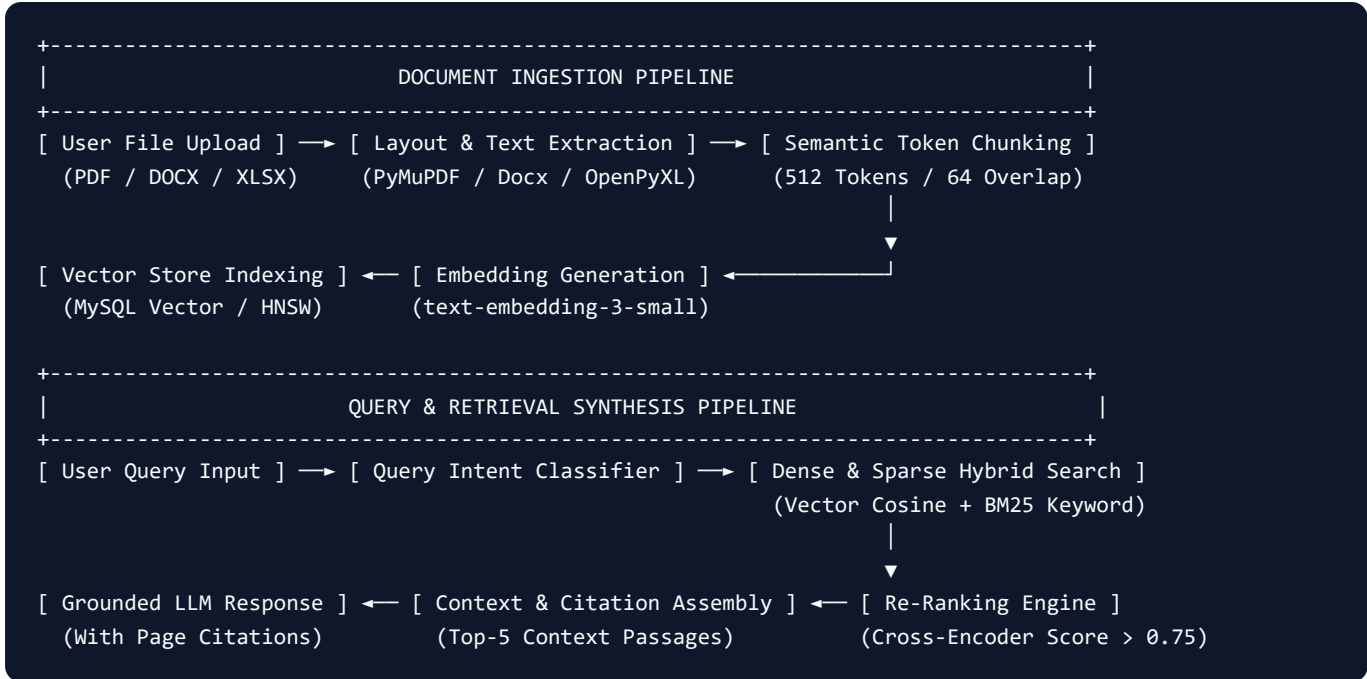
- **Zero Hallucination Grounding:** Enforce strict prompt constraints limiting answers to retrieved passages.
- **Exact Provenance Citation:** Provide page-level and section-level attribution for every generated claim.
- **Multi-Format Ingestion:** Seamless parsing for PDF, DOCX, XLSX, and Markdown files.

Enterprise Governance

- **Multi-Tenancy Isolation:** Metadata pre-filtering guarantees strict `org_id` workspace boundaries.
- **Role-Based Access Control:** Enforces document visibility levels (Executive, Manager, Member).
- **Audit Logging:** Full provenance logging of queries, retrieved vector chunk IDs, and similarity scores.

2. End-to-End Data Flow & Pipeline

The RAG pipeline operates across two decoupled phases: **Document Ingestion & Indexing** (Asynchronous) and **Semantic Retrieval & Synthesis** (Synchronous Execution).



3. Document Ingestion & Extraction Engine

The ingestion engine standardizes unstructured multi-format uploads into standardized, text-searchable chunks with rich structural metadata.

File Format	Parsing Subsystem	Extraction Capabilities
<code>.pdf</code>	PyMuPDF (<code>fitz</code>)	Extracts layout coordinates, multi-column reading flow, embedded tables, and page boundaries.
<code>.docx</code>	<code>python-docx</code>	Parses XML DOM structure, heading hierarchies (H1/H2/H3), paragraph lists, and document properties.
<code>.xlsx</code>	<code>openpyxl</code>	Converts tabular spreadsheet grids into Markdown table representations per worksheet tab.
<code>.txt</code> / <code>.md</code>	Native UTF-8 Stream	Direct plaintext parsing with header boundary preservation.

Chunking & Tokenization Parameters

- **Chunk Strategy:** Semantic Recursive Character Splitter with paragraph boundary alignment.
- **Chunk Target Size:** 512 tokens (~2,000 characters).
- **Chunk Overlap:** 64 tokens (~250 characters) to preserve contextual continuity across chunk boundaries.
- **Metadata Enriched Attributes:** `document_id` , `org_id` , `user_id` , `page_number` , `section_title` , `chunk_index` , `access_role` .

4. Embedding Generation & Vector Storage Schema

Document chunks are converted into high-dimensional dense vector embeddings and indexed for low-latency similarity queries.

⚡ Embedding Configuration

- **Primary Model:** OpenAI `text-embedding-3-small`
- **Vector Dimensions:** 1,536 dimensions
- **Distance Metric:** Cosine Distance (L2 Normalized)
- **Fallback Model:** `bge-small-en-v1.5` (384 dimensions)

🗃 Indexing Strategy

- **Index Type:** HNSW (Hierarchical Navigable Small World)
- **M Parameter:** 16 connections per node
- **efConstruction:** 200 candidate evaluation depth
- **Query Latency Target:** < 15 milliseconds

MySQL Database Schema (`document_embeddings`)

```
CREATE TABLE document_embeddings (  
  id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  document_id BIGINT NOT NULL,  
  org_id INT NOT NULL,  
  user_id INT NOT NULL,  
  chunk_index INT NOT NULL,  
  page_number INT NOT NULL,  
  section_title VARCHAR(255) NULL,  
  chunk_text MEDIUMTEXT NOT NULL,  
  token_count INT NOT NULL,  
  embedding VECTOR(1536) NOT NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  INDEX idx_org_doc (org_id, document_id),  
  VECTOR INDEX idx_embedding (embedding) USING HNSW  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

5. Hybrid Retrieval & Re-Ranking Strategy

To maximize recall accuracy for specialized terminology while preserving semantic search for natural language queries, the system employs a **Hybrid Search & Re-Ranking Pipeline**.

Retrieval Mode	Algorithm / Index	Primary Purpose
Sparse Search	SQL Full-Text (BM25)	Matches exact key phrases, clause numbers, regulation codes (e.g. "ISO-27001 Section 4.2").
Dense Search	Cosine Vector Distance	Captures semantic concepts, synonyms, and intent ("data security requirements").
Score Fusion	Reciprocal Rank Fusion (RRF)	Combines sparse and dense ranks: $RRF_Score(d) = \sum \frac{1}{(60 + Rank(d))}$.
Re-Ranker	Cross-Encoder (<code>bge-reranker-large</code>)	Evaluates Top-20 candidate passages against the query, filtering candidates below 0.75 score.

6. Context Assembly & Grounded Prompt Enforcement

The top 5 re-ranked document passages are formatted into a structured context block injected into the LLM system prompt.

```
SYSTEM PROMPT GUARDRAILS:
You are StratRoom AI Intelligence Hub assistant. Answer the user's inquiry strictly using
the retrieved context passages below.

RULES:
1. Base your answer ONLY on the provided context passages. Do not invent or assume facts.
2. For every assertion or data point, provide an inline citation in the format [Doc: File_Name, Page X].
3. If the context does not contain sufficient details to answer, state: "The uploaded documents do not contain sufficient information to answer your query."

CONTEXT PASSAGES:
[Passage 1 | Doc: Executive_Risk_Report.pdf | Page 4]
The cybersecurity risk index increased by 14% due to delayed patch cycles in legacy servers...

[Passage 2 | Doc: Compliance_Audit.pdf | Page 12]
Section 3.1 requires mandatory multi-factor authentication across all external access portals...
```

7. Multi-Tenancy Isolation & Security Control

- Pre-Retrieval Metadata Filtering:** Vector distance calculations execute *after* strict SQL filtering on `org_id = :current_user_org_id`, ensuring absolute isolation between enterprise tenants.
- Role-Based Access Control (RBAC):** Documents tagged as `executive` visibility are excluded from vector queries initiated by `member` role tokens.
- Path Traversal & Execution Guardrails:** File storage operations enforce realpath canonicalization using `os.path.realpath()` against the root storage directory.

8. Performance Optimization & Cache Architecture



Latency Optimization

- **Embedding Deduplication:** MD5 hashing on chunk content prevents re-embedding identical text passages.
- **Asynchronous Background Chunking:** Upload endpoint returns document handle instantly; chunking executes asynchronously.



Context Compression

- **Dynamic Context Window:** Automatically trims context passages if total prompt length exceeds 4,000 tokens.
- **Semantic Deduplication:** Removes near-duplicate passages with Cosine similarity > 0.92.